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LAB REPORT
On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
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This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by Shaikh Uzair Ahmed **(1BM23CS307)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2024-25. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of Recursive and Non-recursive algorithms using asymptotic notations.
CO2	Apply various design techniques for the given problem.
CO3	Apply the knowledge of complexity classes P, NP, and NP-Complete and prove certain problems are NP-Complete
CO4	Design efficient algorithms and conduct practical experiments to solve problems.

Lab program 1:

Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>

int a[10][10], n, t[10], indegree[10];
int stack[10], top = -1;

void computeIndegree(int, int [][10]);
void tps_SourceRemoval(int, int [][10]);

int main() {
    printf("Enter the no. of nodes: ");
    scanf("%d", &n);
    int i, j;

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    computeIndegree(n, a);
    tps_SourceRemoval(n, a);
    printf("Solution: ");

    for (i = 0; i < n; i++) {
        printf("%d ", t[i]);
    }

    return 0;
}
```

```
void computeIndegree(int n, int a[][10]) {
    int i, j, sum = 0;

    for (i = 0; i < n; i++) {
        sum = 0;
        for (j = 0; j < n; j++) {
            sum = sum + a[j][i];
        }
        indegree[i] = sum;
    }
}

void tps_SourceRemoval(int n, int a[][10]) {
    int i, j, v;

    for (i = 0; i < n; i++) {
        if (indegree[i] == 0) {
            stack[++top] = i;
        }
    }

    int k = 0;
    while (top != -1) {
        v = stack[top--];
        t[k++] = v;

        for (i = 0; i < n; i++) {
            if (a[v][i] != 0) {
                indegree[i] = indegree[i] - 1;
                if (indegree[i] == 0) {
                    stack[++top] = i;
                }
```

```
        }
    }
  }
}
```

```
Enter the no. of nodes: 6
0 0 1 1 0 0
0 0 0 1 1 0
0 0 0 1 0 1
0 0 0 0 0 1
0 0 0 0 0 1
0 0 0 0 0 0
Solution: 1 4 0 2 3 5
```

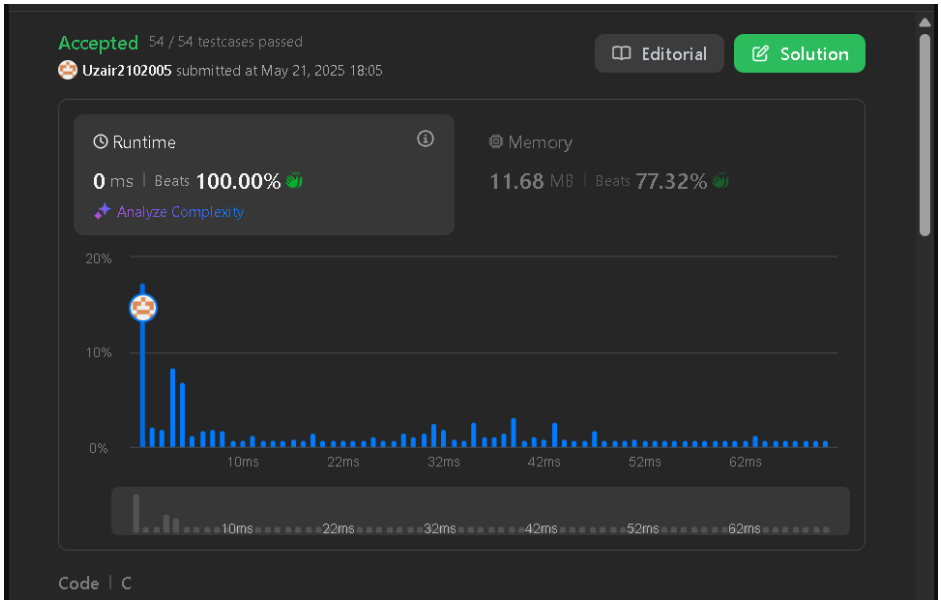
Output

LeetCode 207: Course Schedule

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {

    int* indegree = (int*)calloc(numCourses, sizeof(int));

    int** graph = (int**)calloc(numCourses, sizeof(int*));

    int* graphColSize = (int*)calloc(numCourses, sizeof(int));


    // First pass to count edges to allocate memory

    for (int i = 0; i < prerequisitesSize; i++) {

        int to = prerequisites[i][0];

        graphColSize[prerequisites[i][1]]++;

    }


    // Allocate space for adjacency list

    for (int i = 0; i < numCourses; i++) {

        graph[i] = (int*)malloc(graphColSize[i] * sizeof(int));

        graphColSize[i] = 0; // Reset to use it as an index for inserting

    }
```

```
    // Build graph and indegree array
    for (int i = 0; i < prerequisitesSize; i++) {
        int to = prerequisites[i][0];
        int from = prerequisites[i][1];
        graph[from][graphColSize[from]++] = to;
        indegree[to]++;
    }


    // Initialize queue
    int* queue = (int*)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;


    // Add courses with 0 indegree
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0) {
            queue[rear++] = i;
        }
    }


    int count = 0;


    while (front < rear) {
        int course = queue[front++];
        count++;
        for (int i = 0; i < graphColSize[course]; i++) {
            int neighbor = graph[course][i];
            if (--indegree[neighbor] == 0) {
```

```
                queue[rear++] = neighbor;
            }
        }
    }


    // Cleanup
    for (int i = 0; i < numCourses; i++) {
        free(graph[i]);
    }
    free(graph);
    free(graphColSize);
    free(indegree);
    free(queue);


    return count == numCourses;
}
```



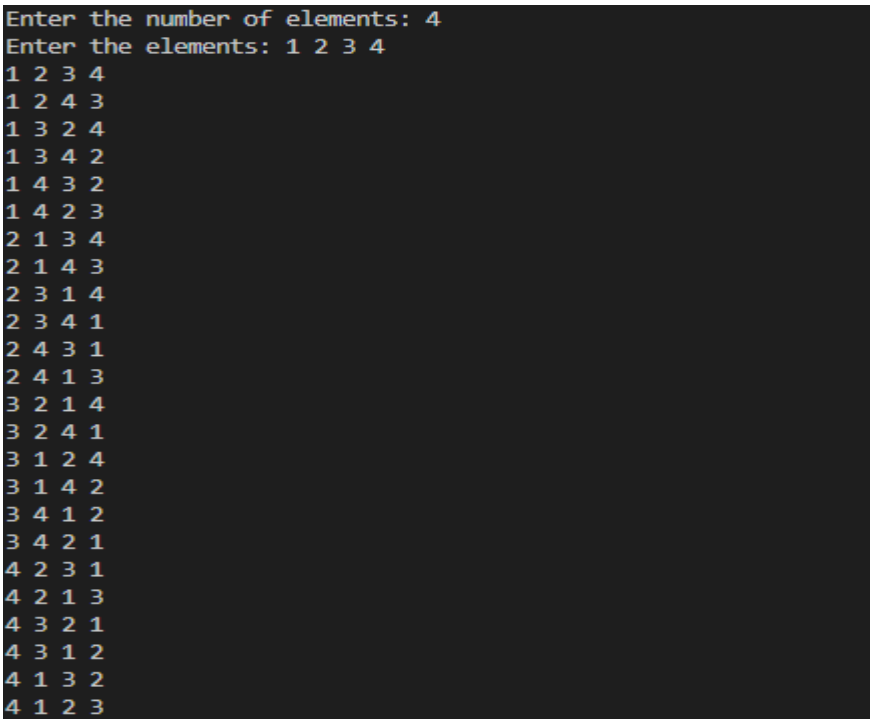
Output

Lab program 2:

Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>
#include <stdlib.h>

void swap(int* a, int* b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

void generatePermutations(int arr[], int start, int end) {
    if (start == end) {
        for (int i = 0; i <= end; i++) {
            printf("%d ", arr[i]);
        }
        printf("\n");
    } else {
        for (int i = start; i <= end; i++) {
            swap(&arr[start], &arr[i]);
            generatePermutations(arr, start + 1, end);
            swap(&arr[start], &arr[i]); // backtrack
        }
    }
}

int main() {
    int n;
    printf("Enter the number of elements: ");
```

```
    scanf("%d", &n);

    int* arr = (int*)malloc(n * sizeof(int));
    printf("Enter the elements: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    generatePermutations(arr, 0, n - 1);
    free(arr);

    return 0;
}
```



Output

Lab program 3:

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>


int *a;


void mergeSort(int a[], int low, int high);

void merge(int a[], int low, int high, int mid);


int main() {

   int n;

   printf("Enter Number of Elements");

   scanf("%d", &n);

   a = (int*)malloc(sizeof(int) * n);



   for (int i = 0; i < n; ++i) {

      a[i]=rand();

   }



   clock_t start = clock();

   mergeSort(a, 0, n - 1);

   clock_t end = clock();



   double t = ((double)(end - start)) / CLOCKS_PER_SEC;
```

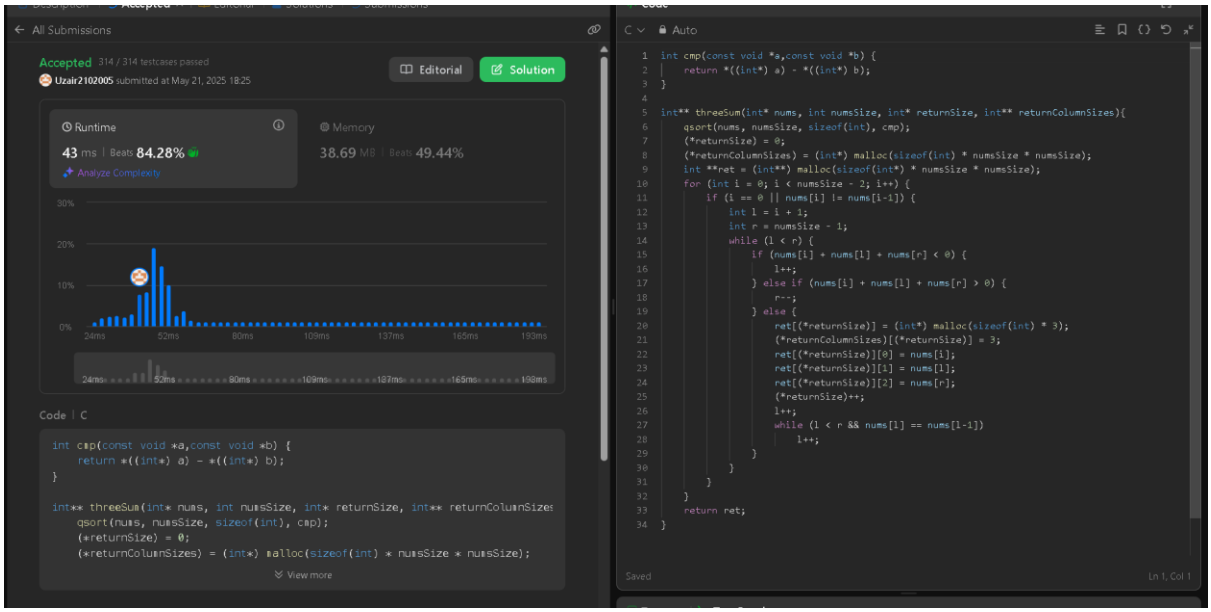
```
    printf("Sorted Array\n");
    for (int i = 0; i < n; ++i) {
        printf("%d ", a[i]);
    }
    printf("\n");


    printf("%.30f\n", t);


    free(a);

    return 0;
}


void mergeSort(int a[], int low, int high) {
    if (low < high) {
        int mid = (low + high) / 2;
        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);
        merge(a, low, high, mid);
    }
}


void merge(int a[], int low, int high, int mid) {
    int *tmp = (int*)malloc((high - low + 1) * sizeof(int));
    int i = low, j = mid + 1, k = 0;
```

```
    while (i <= mid && j <= high) {
        if (a[i] < a[j]) {
            tmp[k++] = a[i++];
        } else {
            tmp[k++] = a[j++];
        }
    }

    while (i <= mid) {
        tmp[k++] = a[i++];
    }


    while (j <= high) {
        tmp[k++] = a[j++];
    }


    for (int i = 0; i < (high - low + 1); ++i) {
        a[low + i] = tmp[i];
    }

    free(tmp);
}
```

```
Enter Number of Elements10
Sorted Array
424238335 596516649 719885386 846930886 1189641421 1649760492 1681692777
    1714636915 1804289383 1957747793
0.000003999999999999999818992447


=== Code Execution Successful ===
```

Output

LeetCode 15: 3Sum



Output

Lab program 4:

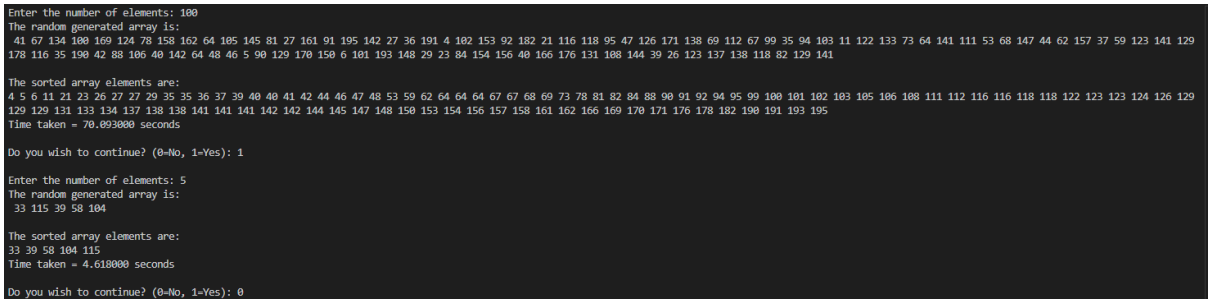
Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <time.h>
#include <stdio.h>
#include <stdlib.h>  // for rand()
#include <windows.h> // for Sleep()

#define MAX 5000

void quicksort(int[], int, int);
int partition(int[], int, int);

int main()
{
    int i, n, a[MAX], ch = 1;  // Initialize ch = 1
    clock_t start, end;
```

```
    while(ch)
    {
        printf("\nEnter the number of elements: ");
        scanf("%d", &n);

        for(i = 0; i < n; i++)
            a[i] = rand() % 200;

        printf("The random generated array is:\n");
        for(i = 0; i < n; i++)
            printf(" %d", a[i]);

        start = clock();
        quicksort(a, 0, n - 1);
        end = clock();

        printf("\n\nThe sorted array elements are:\n");
        for(i = 0; i < n; i++)
            printf("%d ", a[i]);

        printf("\nTime taken = %f seconds", (double)(end - start) / CLOCKS_PER_SEC);
        printf("\n\nDo you wish to continue? (0=No, 1=Yes): ");
        scanf("%d", &ch);
    }
    return 0;
}

void quicksort(int a[], int low, int high)
{
```

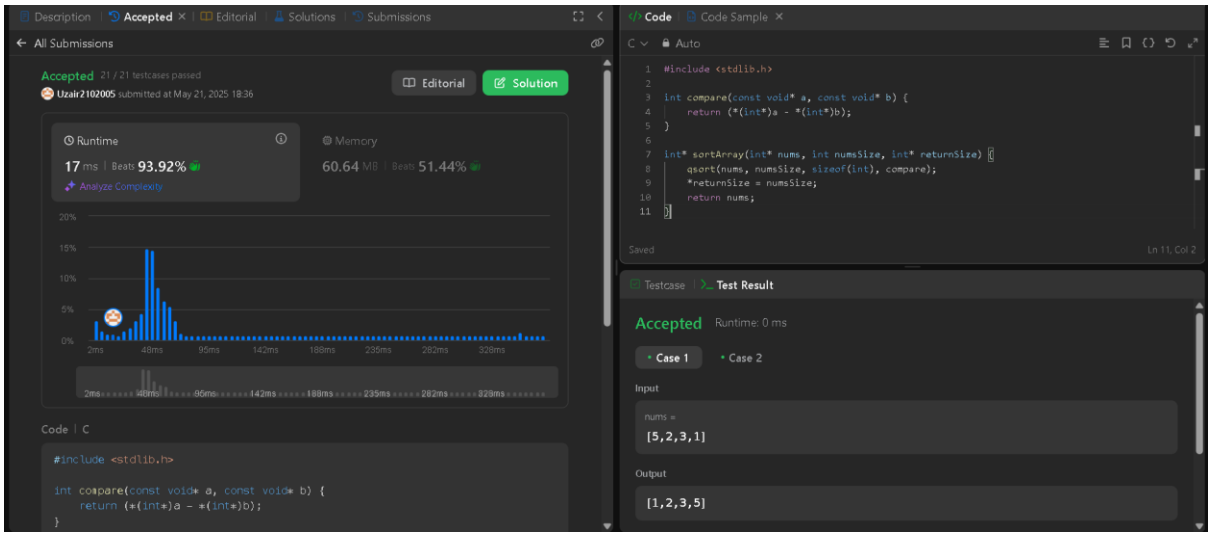
```
    int mid;
    Sleep(500); // Windows-specific delay (500ms)
    if(low < high)
    {
        mid = partition(a, low, high);
        quicksort(a, low, mid - 1);
        quicksort(a, mid + 1, high);
    }
}

int partition(int a[], int low, int high)
{
    int key = a[low];
    int i = low + 1;
    int j = high;

    while(i <= j)
    {
        while(i <= high && key >= a[i])
            i++;
        while(key < a[j])
            j--;

        if(i < j)
        {
            int temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
```

```
        else
        {
            int temp = a[j];
            a[j] = a[low];
            a[low] = temp;
        }
    }
    return j;
}
```



Output

LeetCode 912: Sort an Array



Output

Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include <stdio.h>
#include <time.h>

void heapify(int a[], int n, int i) {
    int largest = i;
    int left = 2*i + 1;
    int right = 2*i + 2;
    int temp;

    if (left < n && a[left] > a[largest])
        largest = left;

    if (right < n && a[right] > a[largest])
        largest = right;

    if (largest != i) {
        temp = a[i];
        a[i] = a[largest];
        a[largest] = temp;

        heapify(a, n, largest);
    }
}

void heapSort(int a[], int n) {
```

```
    int temp;
    for (int i = n / 2 - 1; i >= 0; i--) {
        heapify(a, n, i);
    }

    for (int i = n - 1; i >= 0; i--) {
        temp = a[0];
        a[0] = a[i];
        a[i] = temp;

        heapify(a, i, 0);
    }
}

int main() {
    int n, a[20], ch = 1;
    clock_t start, end;

    while (ch) {
        printf("\nEnter the number of elements to sort: ");
        scanf("%d", &n);

        printf("Enter the elements to sort: ");
        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }
```

```
        start = clock();
        heapSort(a, n);
        end = clock();

        printf("\nThe sorted list of elements is:\n");
        for (int i = 0; i < n; i++) {
            printf("%d\n", a[i]);
        }

        printf("\nTime taken is %lf seconds\n", (double)(end - start) / CLOCKS_PER_SEC);

        printf("Do you wish to run again (0/1)? ");
        scanf("%d", &ch);
    }

    return 0;
}
```

```
Enter the number of elements to sort: 5
Enter the elements to sort:  1 4 2 3 7

The sorted list of elements is:
1
2
3
4
7

Time taken is 0.000000 seconds
Do you wish to run again (0/1)? 0
```

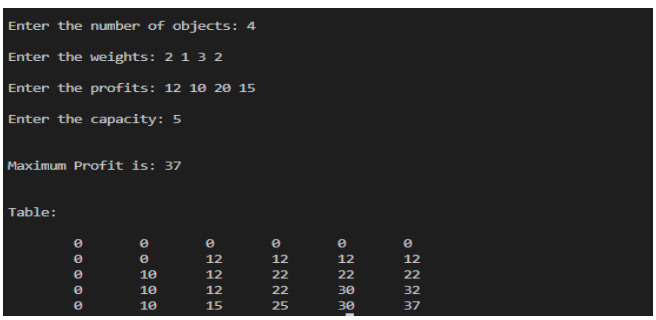
Output

Lab program 6:

Implement 0/1 Knapsack problem using dynamic programming.

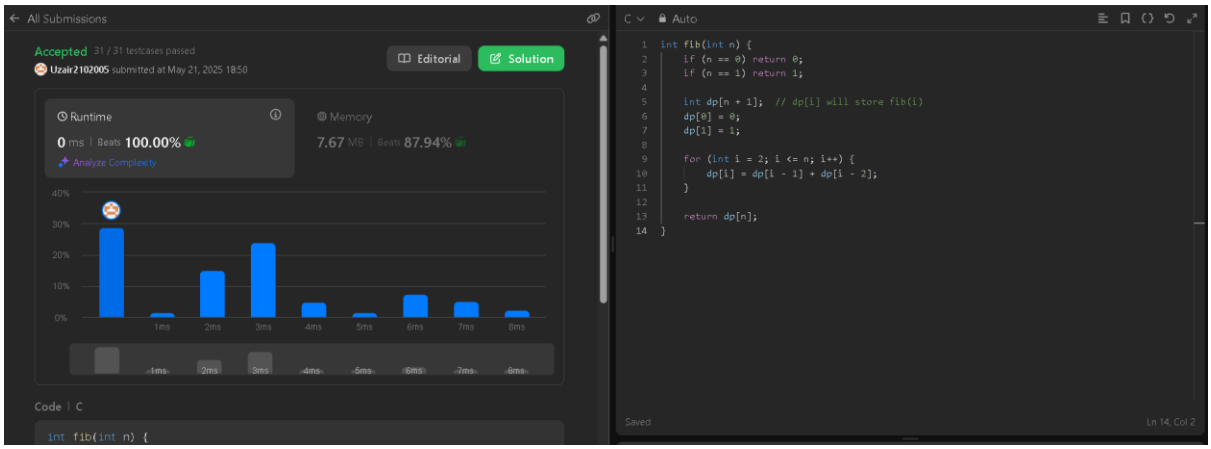
```
#include <stdio.h>

int i, j, n, c, w[10], p[10], v[10][10];


int max(int a, int b) {
   return (a > b) ? a : b;
}
void knapsack(int n, int w[10], int p[10], int c) {
   for (i = 0; i <= n; i++) {
      for (j = 0; j <= c; j++) {
         if (i == 0 || j == 0)
            v[i][j] = 0;
         else if (w[i] > j)
            v[i][j] = v[i - 1][j];
         else
            v[i][j] = max(v[i - 1][j], v[i - 1][j - w[i]] + p[i]);
      }
   }
   printf("\n\nMaximum Profit is: %d\n", v[n][c]);
   printf("\n\nTable:\n\n");
   for (i = 0; i <= n; i++) {
      for (j = 0; j <= c; j++) {
         printf("\t%d", v[i][j]);
      }
      printf("\n");
   }
```

```
}

void main() {
   printf("\nEnter the number of objects: ");
   scanf("%d", &n);

   printf("\nEnter the weights: ");
   for (i = 1; i <= n; i++) {
      scanf("%d", &w[i]);
   }

   printf("\nEnter the profits: ");
   for (i = 1; i <= n; i++) {
      scanf("%d", &p[i]);
   }

   printf("\nEnter the capacity: ");
   scanf("%d", &c);

   knapsack(n, w, p, c);
}
```



Output

LeetCode 509: Fibonacci Number



Output

Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include <stdio.h>


int n;
int a[10][10];
int D[10][10];
void write_data() {
    int i, j;
    printf("The Distance matrix is shown below\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            printf("%d", D[i][j]);
            printf(" ");
        }
        printf("\n");
    }
}
```

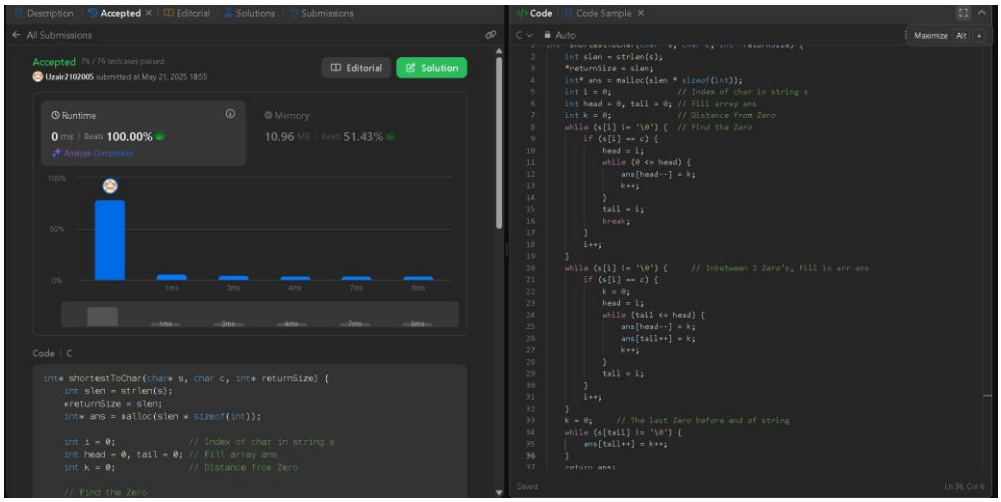
```
void read_data() {
    int i, j;
    printf("Enter the no of nodes\n");
    scanf("%d", &n);
    printf("Enter the adjacency matrix\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &a[i][j]);
        }
    }

}
int min(int a, int b) {
    if (a < b) {
        return a;
    } else {
        return b;
    }
}
void Floyd() {
    int i, j, k;
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            D[i][j] = a[i][j];
        }
    }
```

```
    for (k = 0; k < n; k++) {
        for (i = 0; i < n; i++) {
            for (j = 0; j < n; j++) {
                D[i][j]=min(D[i][j],(D[i][k]+D[k][j]));
            }
        }
    }
}
int main() {
    read_data();
    Floyd();
    write_data();
    return 0;
}
```

```
Enter the no of nodes
4
Enter the adjacency matrix
0 99 3 99
2 0 99 99
99 6 0 1
7 99 99 0
The Distance matrix is shown below
0 9 3 4
2 0 5 6
8 6 0 1
7 16 10 0
```

Output

LeetCode 821: Shortest Distance to a Character



Output

Lab program 8:

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>

int cost[10][10], n, t[10][2], sum;

void prims(int cost[10][10], int n);

int main() {
    int i, j;

    printf("Enter the number of vertices: ");
    scanf("%d", &n);

    printf("Enter the cost adjacency matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &cost[i][j]);
```

```
        }
    }

    prims(cost, n);

    printf("Edges of the minimal spanning tree:\n");
    for (i = 0; i < n - 1; i++) {
        printf("(%d, %d) ", t[i][0], t[i][1]);
    }
    printf("\nSum of minimal spanning tree: %d\n", sum);

    return 0;
}

void prims(int cost[10][10], int n) {
    int i, j, u, v;
    int min, source;
    int p[10], d[10], s[10];

    min = 999;
    source = 0;

    // Initialize arrays
    for (i = 0; i < n; i++) {
        d[i] = cost[source][i];
        s[i] = 0;
        p[i] = source;
```

```
}
s[source] = 1;
sum = 0;
int k = 0;

// Find MST
for (i = 0; i < n - 1; i++) {
   min = 999;
   u = -1;

   // Find the vertex with minimum distance to the MST
   for (j = 0; j < n; j++) {
      if (s[j] == 0 && d[j] < min) {
         min = d[j];
         u = j;
      }
   }

   if (u != -1) {
      // Add edge to MST
      t[k][0] = u;
      t[k][1] = p[u];
      k++;
      sum += cost[u][p[u]];
      s[u] = 1;

      // Update distances
```

```
        for (v = 0; v < n; v++) {
            if (s[v] == 0 && cost[u][v] < d[v]) {
                d[v] = cost[u][v];
                p[v] = u;
            }
        }
    }
  }
}
```

```
Enter the number of vertices: 4
Enter the cost adjacency matrix:
0 1 5 2
1 0 99 99
5 99 0 3
2 99 3 0
Edges of the minimal spanning tree:
(1, 0) (3, 0) (2, 3)
Sum of minimal spanning tree: 6
```

Output

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>

int cost[10][10], n, t[10][2], sum;

void kruskal(int cost[10][10], int n);
int find(int parent[10], int i);

int main() {
   int i, j;
```

```
    printf("Enter the number of vertices: ");
    scanf("%d", &n);

    printf("Enter the cost adjacency matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &cost[i][j]);
        }
    }

    kruskal(cost, n);

    printf("Edges of the minimal spanning tree:\n");
    for (i = 0; i < n - 1; i++) {
        printf("(%d, %d) ", t[i][0], t[i][1]);
    }
    printf("\nSum of minimal spanning tree: %d\n", sum);

    return 0;
}

void kruskal(int cost[10][10], int n) {
    int min, u, v, count, k;
    int parent[10];

    k = 0;
    sum = 0;
```

```
// Initialize parent array for Union-Find
for (int i = 0; i < n; i++) {
    parent[i] = i;
}

count = 0;
while (count < n - 1) {
    min = 999;
    u = -1;
    v = -1;

    // Find the minimum edge
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (find(parent, i) != find(parent, j) && cost[i][j] < min) {
                min = cost[i][j];
                u = i;
                v = j;
            }
        }
    }

    // Perform Union operation
    int root_u = find(parent, u);
    int root_v = find(parent, v);
```

```
        if (root_u != root_v) {
            parent[root_u] = root_v;
            t[k][0] = u;
            t[k][1] = v;
            sum += min;
            k++;
            count++;
        }
    }
}

int find(int parent[10], int i) {
    while (parent[i] != i) {
        i = parent[i];
    }
    return i;
}
```

```
PS C:\Users\STUDENT\Desktop\KRUSKALS> cd "c:\Users\STUDENT\Desktop\KRUSKALS\" ; if ($?) { gcc kruskals.c -o kruskals } ; if ($?) { .\kruskals }
Enter the number of vertices: 4
Enter the cost adjacency matrix:
0 1 5 2
1 0 99 99
5 99 0 3
2 99 3 0
Edges of the minimal spanning tree:
(0, 1) (0, 3) (2, 3)
Sum of minimal spanning tree: 6
PS C:\Users\STUDENT\Desktop\KRUSKALS>
```

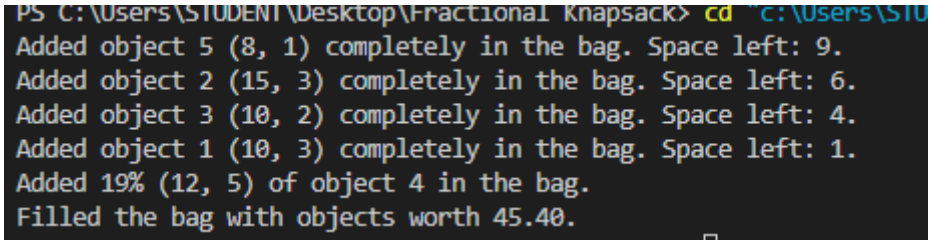
Output

Lab program 9:

Implement Fractional Knapsack using Greedy technique.

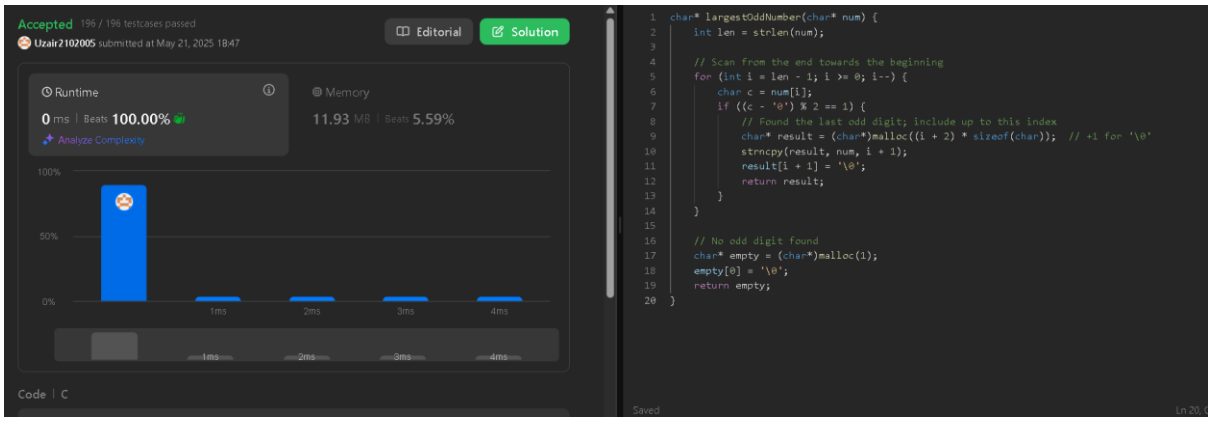
```
#include <stdio.h>


int n = 5;

int p[10] = {3, 3, 2, 5, 1}; /*Weight of items*/

int w[10] = {10, 15, 10, 12, 8};/*Profit of items*/

int W = 10;


int main() {

   int cur_w;

   float tot_v = 0;

   int i, maxi;

   int used[10];


   for (i = 0; i < n; ++i)

      used[i] = 0;


   cur_w = W;


   while (cur_w > 0) {

      maxi = -1;


      for (i = 0; i < n; ++i)

         if ((used[i] == 0) && ((maxi == -1) || ((float)w[i] / p[i] > (float)w[maxi] / p[maxi])))

            maxi = i;


      used[maxi] = 1;
```

```
        cur_w -= p[maxi];
        tot_v += w[maxi];

        if (cur_w >= 0)
            printf("Added object %d (%d, %d) completely in the bag. Space left: %d.\n", maxi + 1, w[maxi], p[maxi], cur_w);
        else {
            printf("Added %d%% (%d, %d) of object %d in the bag.\n", (int)((1 + (float)cur_w / p[maxi]) * 100), w[maxi], p[maxi], maxi + 1);
            tot_v -= w[maxi];
            tot_v += (1 + (float)cur_w / p[maxi]) * w[maxi];
        }
    }

    printf("Filled the bag with objects worth %.2f.\n", tot_v);
    return 0;
}
```



Output

LeetCode 1903: Largest Odd Number in a String



Output

Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>


void main() {
    int i, j, n, v, k, min, u, c[20][20], s[20], d[20];



    printf("\nEnter the number of vertices: ");
    scanf("%d", &n);


    printf("\nEnter the cost adjacency matrix:");
    printf("\n(Enter 999 for no edge)\n");
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            scanf("%d", &c[i][j]);
        }
```

```
}

printf("\nEnter the source vertex: ");
scanf("%d", &v);

// Initialize distances and visited set
for (i = 1; i <= n; i++) {
    s[i] = 0;
    d[i] = c[v][i];
}

d[v] = 0;
s[v] = 1;

for (k = 2; k <= n; k++) {
    min = 999;

    // Find the minimum distance vertex from the set of vertices not yet processed
    for (i = 1; i <= n; i++) {
        if ((s[i] == 0) && (d[i] < min)) {
            min = d[i];
            u = i;
        }
    }

    s[u] = 1;
```

```
        // Update distance value of the adjacent vertices
        for (i = 1; i <= n; i++) {
            if (s[i] == 0) {
                if (d[i] > (d[u] + c[u][i])) {
                    d[i] = d[u] + c[u][i];
                }
            }
        }
    }

    printf("\nThe shortest distances from vertex %d are:\n", v);
    for (i = 1; i <= n; i++) {
        printf("%d --> %d = %d\n", v, i, d[i]);
    }

}
```

```
Enter the number of vertices: 5

Enter the cost adjacency matrix:
(Enter 999 for no edge)
999 7 3 999 999
7 999 2 5 4
3 2 999 4 999
999 5 4 999 6
999 4 999 6 999

Enter the source vertex: 2

The shortest distances from vertex 2 are:
2 --> 1 = 5
2 --> 2 = 0
2 --> 3 = 2
2 --> 4 = 5
2 --> 5 = 4
```

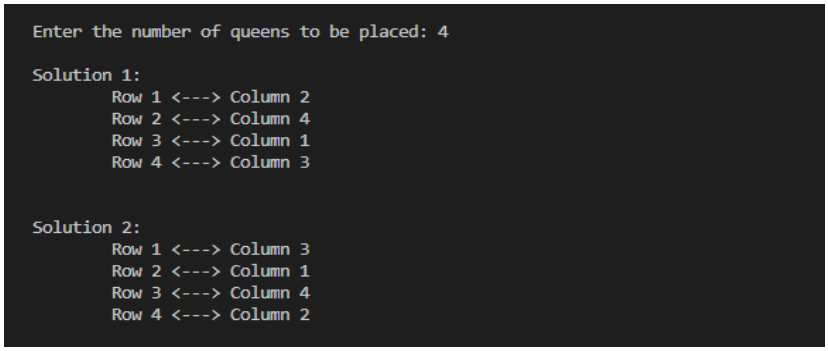
Output

Lab program 11:

Implement “N-Queens Problem” using Backtracking.

```
#include <stdio.h>
#include <math.h>

int x[20], count = 1;  // Array to store the positions of the queens

void queens(int k, int n);
int place(int k, int j);

int main() {
    int n, k = 1;
    printf("\nEnter the number of queens to be placed: ");
    scanf("%d", &n);
    queens(k, n);
    return 0;
}

void queens(int k, int n) {
    int j;
    for (j = 1; j <= n; j++) {
        if (place(k, j)) {
            x[k] = j;  // Place the queen in the current position
            if (k == n) {  // If all queens are placed
                printf("\nSolution %d:\n", count);
                count++;
                for (int i = 1; i <= n; i++) {
```

```
                printf("\tRow %d <---> Column %d\n", i, x[i]);
            }
            printf("\n");
        } else {
            queens(k + 1, n);  // Try to place the next queen
        }
    }
  }
}

int place(int k, int j) {
  for (int i = 1; i < k; i++) {
    // Check if the queen is in the same column or on the same diagonal
    if (x[i] == j || abs(x[i] - j) == abs(i - k)) {
      return 0;  // Conflict detected
    }
  }
  return 1;  // No conflict, safe to place the queen
}
```



Output